Credit Default Swaps

Seminar Module

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Topics Covered

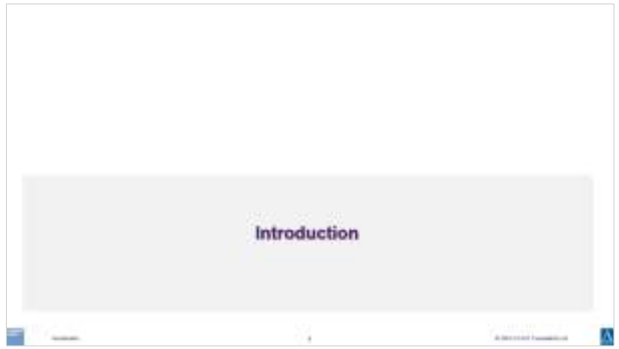
- Introduction
  - Definition of a Credit Default Swap (CDS)
  - Credit events
  - Settlement methods
- Market conventions
- Recovery products
- Recent events
- How auction settlement works
- Introduction to CDS pricing

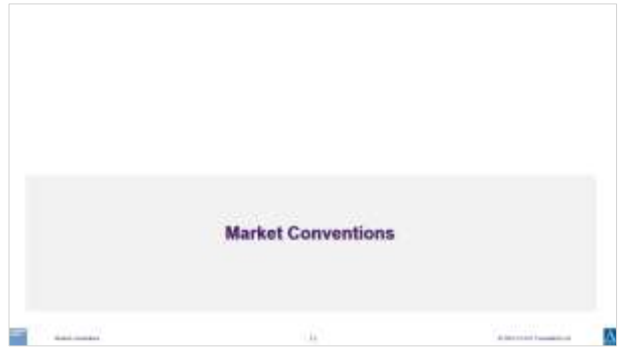
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Key Point Summary

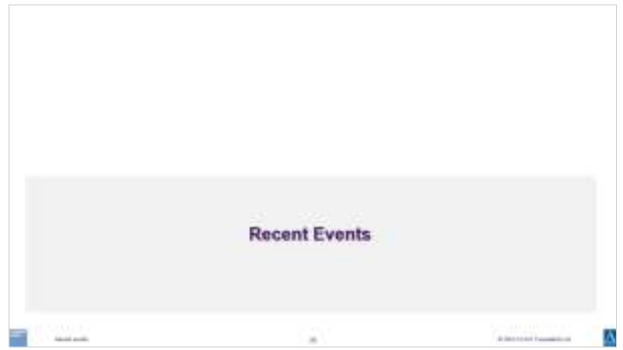
- CDS premiums are normally paid quarterly on Adjusted CDS Dates
- CDS premiums are now standardised, mostly at 100bp p.a. and 500bp p.a., with an upfront payment
- The upfront payment adjusts for the standardised premium being different from the market quote, and for premium accrual
- Upfront payments are calculated by using the ISDA CDS Standard Model
- CDS settlements are normally through cash settlement following an auction
- Five Determination Committees are responsible for making key decisions on specific credit events and reference entities
- Recovery products like Recovery Lock allow users to hedge or speculate on the recovery rate following a credit event

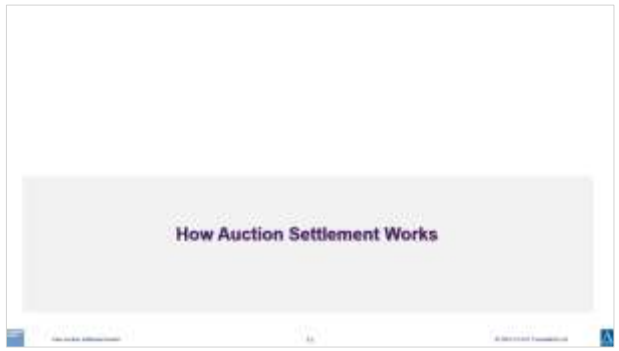
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Introduction

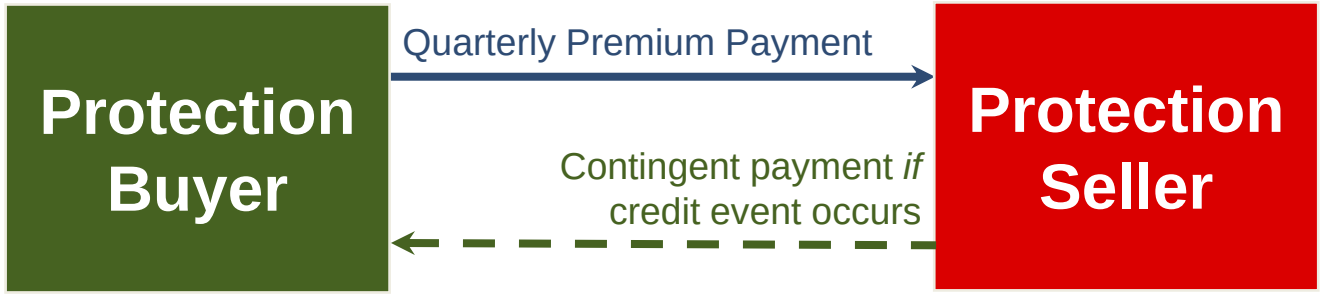
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Credit Default Swaps

A Credit Default Swap (CDS) involves two counterparties:

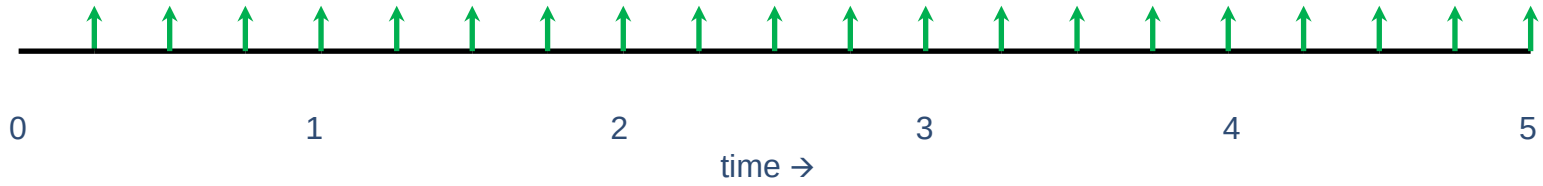
- The Protection Buyer:
  - Pays an agreed fee
  - Receives a contingent payment if a credit event occurs to the reference entity
- The Protection Seller:
  - Receives an agreed fee
  - Makes a contingent payment if a credit event occurs to the reference entity

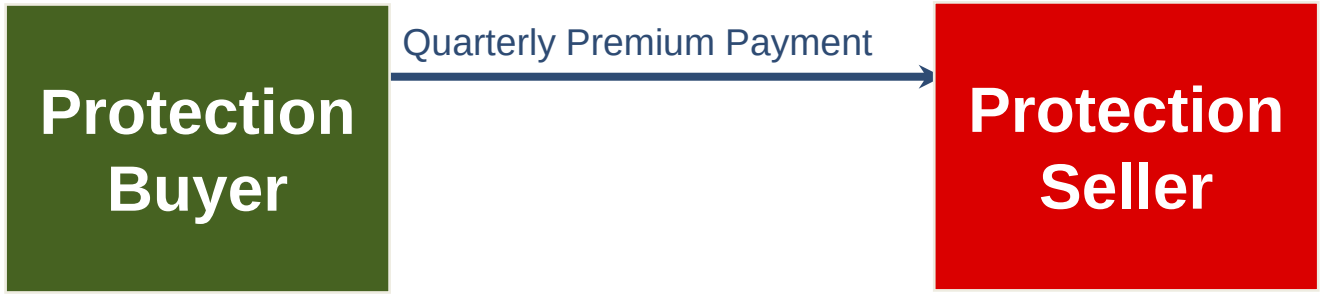
- The fee (a.k.a. premium or spread or coupon) is currently paid quarterly
  - Expressed in basis points per annum on an Act/360 basis
  - Payments are made on the 20th of respectively March, June, September or December
  - There may also be an upfront fee



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5 Year CDS – No Credit Event

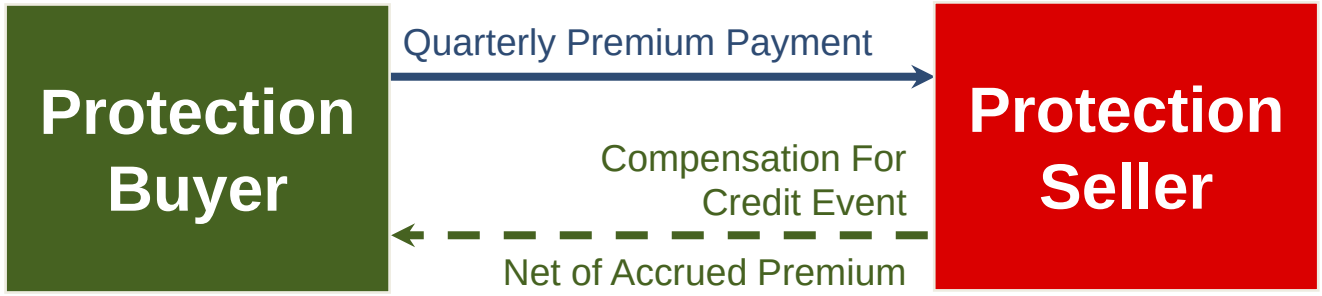




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5 Year CDS – Credit Event





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Effect of CDS on Each Party

The Protection Buyer:

- Effectively pays an insurance premium
- Has purchased a hedge against credit risk
- Is ‘short’ credit risk

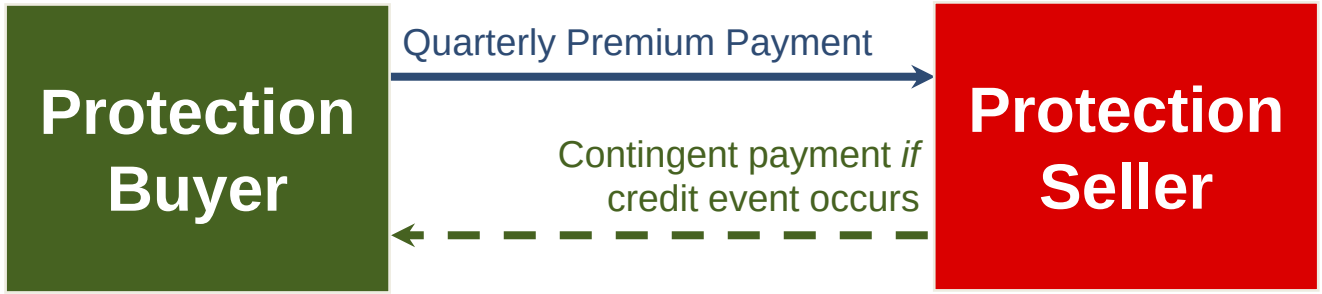
The Protection Seller:

- Earns the premium
- Takes an off-balance sheet credit exposure
- Is ‘long’ credit risk

An important difference between CDS contracts and insurance contracts is that:

CDS contracts have value

As market conditions change, the value of an existing CDS contract can rise or fall.



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Terms and Definitions

- Reference Entity
  - The legal entity which borrows money
  - May be a sovereign, financial institution, corporation, or another specified entity
- Reference Obligation
  - Specifies the level of subordination within the reference entity's debt structure for determining:
    - Which bonds can be included in the auction if 'Auction Settlement' takes place following a credit event
    - What can be delivered if 'Physical Delivery' takes place following a credit event
    - Standard Reference Obligation introduced in the 2014 Credit Derivatives Definitions
- Notional Principal
  - The hypothetical quantity upon which premiums and other payment obligations are computed
- Dates
  - **Trade Date** – the date on which the parties agree to the terms of a contract
  - **Effective Date** – the date on which the parties begin calculating accrued obligations
  - **Termination Date** – the date on which the CDS expires
- Credit Event
  - An event that happens in respect of the reference entity that triggers payment under the CDS
- Settlement Method
  - The way in which the protection buyer is compensated after the occurrence of a credit event

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Credit Events

What defines a ‘credit event’ is covered by ISDA 2014 terms, and includes:

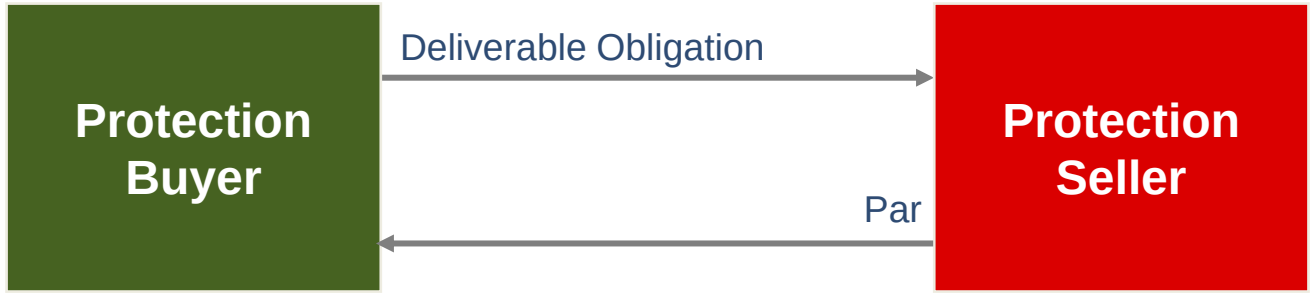
- Bankruptcy
- Failure to pay
  - Borrowed money
  - Payment more than $1m
  - After pre-specified grace period
- Restructuring (MR, MM, CR/FR – instead of NR/XR)
  - Reduction or postponement of interest or principal repayments
  - Change in obligation’s seniority causing obligation to become subordinated
  - Change in currency or composition of material debt obligation…
  - … in circumstances where the financial condition of the Reference Entity has deteriorated
- Obligation acceleration
- Obligation default (e.g. violation of a covenant but not a failure to pay)
- Repudiation or moratorium (sovereigns)
- Government Intervention (financials)

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Settlement Methods – Physical Delivery

Historically, settlement following a credit event was made by **Physical Delivery**:

Payment of par in exchange for receipt of 'deliverable obligation'



Deliverable Obligations can be:

- Direct obligations of the reference entity so long as they rank pari passu with reference obligation (or better)
- Obligations of a subsidiary of the reference entity
- Obligations of a third party guaranteed by the reference entity

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Settlement Methods – Cash Settlement

As an alternative to Physical Delivery, contracts could specify **Cash Settlement**

- ... payment by the protection seller of a cash amount equal to:

Par [or Initial Price] **less** Post-default Market Value

- Post-default Market Value established after 14-30 days after credit event by…
  - Quote service
  - Dealer poll (usually 3-5 dealers)

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Settlement Methods – Auction Settlement

- The ISDA auction methodology has developed over several years
  - Prior to 2009 it required an ‘opt-in’ protocol or trades would be settled bilaterally
  - Following the ‘2009 Supplement’, **Auction Settlement** is now the market standard
- Objective: to determine a fair cash settlement price for CDS
- Following a credit event, the Credit Derivatives Determination Committee (DC):
  - Decides if both a senior and a subordinated auction is required
  - Publishes ‘Auction Terms’, including timetable, auction deliverables etc.
  - Does not **have** to call an auction
- Auction itself is administered by Markit and Creditex as a single-price auction with two-steps:
  - Initial Auction
  - Final Auction

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Settlement Procedures

Once a credit event has occurred, there are 60 days to begin CDS settlement:

- Event Determination Date
  - Usually is the date on which the Credit Event Resolution Request was submitted to the DC
- The Credit Derivatives Determination Committee (DC) manages the auction process

- Fallback to Physical Settlement remains available:
  - When DC does not choose to hold an auction*, and for US Municipal CDS
  - Within 30 days of the Event Determination Date (ignoring any days the DC took to deliberate), the buyer must serve a Notice of Physical Settlement ("NOPS") to the seller
    - The NOPS is a confirmation of the buyer's delivery plans in settlement
    - It includes a detailed description of the deliverable obligation or obligations
    - The buyer can, on or before the Physical Settlement Date, notify the seller that it will change the obligations to be delivered…
    - … This is the **Cheapest To Deliver Option** (CTD option)

** as with Irish Bank Resolution Corporation credit event on 7th Feb 2013*

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Credit Derivatives Determinations Committees

- There is a Determination Committee (DC) for each of the following regions:
  - Americas
  - EMEA (Europe)
  - Japan
  - Asia excluding Japan
  - Australia & New Zealand
- Each DC is responsible for resolving issues in their region pertaining to:
  - Credit Events
  - CDS Auctions
  - Succession Events
  - Substitute Reference Obligations
- Membership of DCs:
  - 8-9 global dealers
  - 3 global non-dealer ISDA members (at least $1bn AuM, and CDS exposure)
  - 2 consultative (non-voting) CCPs
- Administered by ICE Benchmark Administration

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Determinations Committees Membership

Voting Members

- 9 Dealers
  - Bank of America
  - Barclays
  - BNP Paribas
  - Citibank
  - Credit Suisse
  - Deutsche Bank
  - Goldman Sachs
  - JPMorgan Chase Bank
  - Mizuho Securities*
- 3 Non-Dealers
  - Citadel LLC
  - Elliott Management Corporation
  - PIMCO LLC

Consultative Members (not all regions)

- 2 CCPs
  - ICE Clear Credit / ICE Clear Europe
  - LCH

* except for Australia/NZ

Source: CDSdeterminationscommittees.org, 29 Apr 2023

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Market Conventions

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Buying and Selling

Buying protection

- Pay premium
- Receive settlement after credit event
- Sells risk
- Shorts the credit
- Equivalent to selling a bond

Selling protection

- Receive premium
- Pay settlement after credit event
- Buys risk
- Goes long the credit
- Equivalent to buying a bond

2014 ISDA Credit Derivatives Definitions

- **Buyer** means the Fixed Rate Payer
  - i.e. the protection buyer
- **Seller** means the Floating Rate Payer
  - i.e. the protection seller

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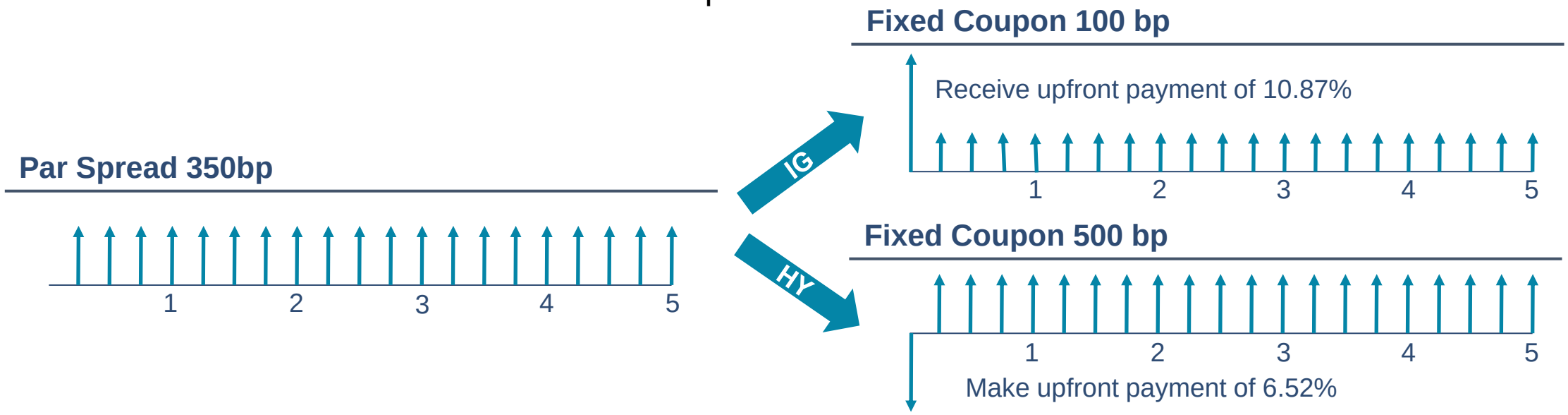
Market Conventions

- Quotations are in basis points per annum
  - e.g. 160 is 160 bp p.a. or 1.60% p.a.
- Coupons:
  - Coupon calculated using Act/360 convention
  - Paid quarterly in arrears on Adjusted CDS Dates
  - Effective Date is T+1 for premium accrual
- Dates:
  - **CDS Dates** are currently Mar 20, Jun 20, Sep 20, and Dec 20 each year
  - **Adjusted CDS Dates** are CDS dates adjusted using the following business day convention
- Maturity
  - Standard maturity is five years…
  - … rounded to nearest half-year date (20th June and 20th December)

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Standardised Premiums

- Standardised premiums have applied since 2009:
  - For most countries, premiums are fixed at 100 bp p.a. (inv. grade), or 500 bp p.a. (high-yield)
  - Accrual start date from (and including) the previous adjusted CDS date, NOT trade effective date
  - An **up-front payment** then compensates for:
    - Difference between 100 bp (or 500bp) running premium and current market quote
    - Premium accrued since last quarterly payment date
- The three sets of cash flows below are all equivalent:



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Standardised Premiums – Who Pays Whom?

The ‘off-market’ adjustment

- Who pays whom depends upon whether the market quote (aka the ‘par spread’) is higher or lower than the standardised premium:

	Market < Standard	Market > Standard
Buyer	Receives	Pays
Seller	Pays	Receives

The accrued premium

Buyer	Receives
Seller	Pays

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Standardised Premiums and Recovery Rates

- Standardised premiums were introduced region-by-region in 2009:

		Coupon Strikes			
	Implementation Date	25	100	500	1000
Asian Corporate & Sovereign CDS	December, 2009		☑	☑	
Australian Corporate & Sovereign CDS	September, 2009		☑	☑	
Eastern & Central European Corporate & Sovereign CDS	September, 2009		☑	☑	
European Corporate CDS	June, 2009	☑	☑	☑	☑
Japanese Corporate & Sovereign CDS	December, 2009	☑	☑	☑	
Latin American Corporate & Sovereign CDS	September, 2009		☑	☑	
Middle Eastern Corporate & Sovereign CDS	September, 2009		☑	☑	
New Zealand Corporate & Sovereign CDS	September, 2009		☑	☑	
North American Corporate & Sovereign CDS	April, 2009		☑	☑	
Western European Sovereign CDS	June, 2009	☑	☑		

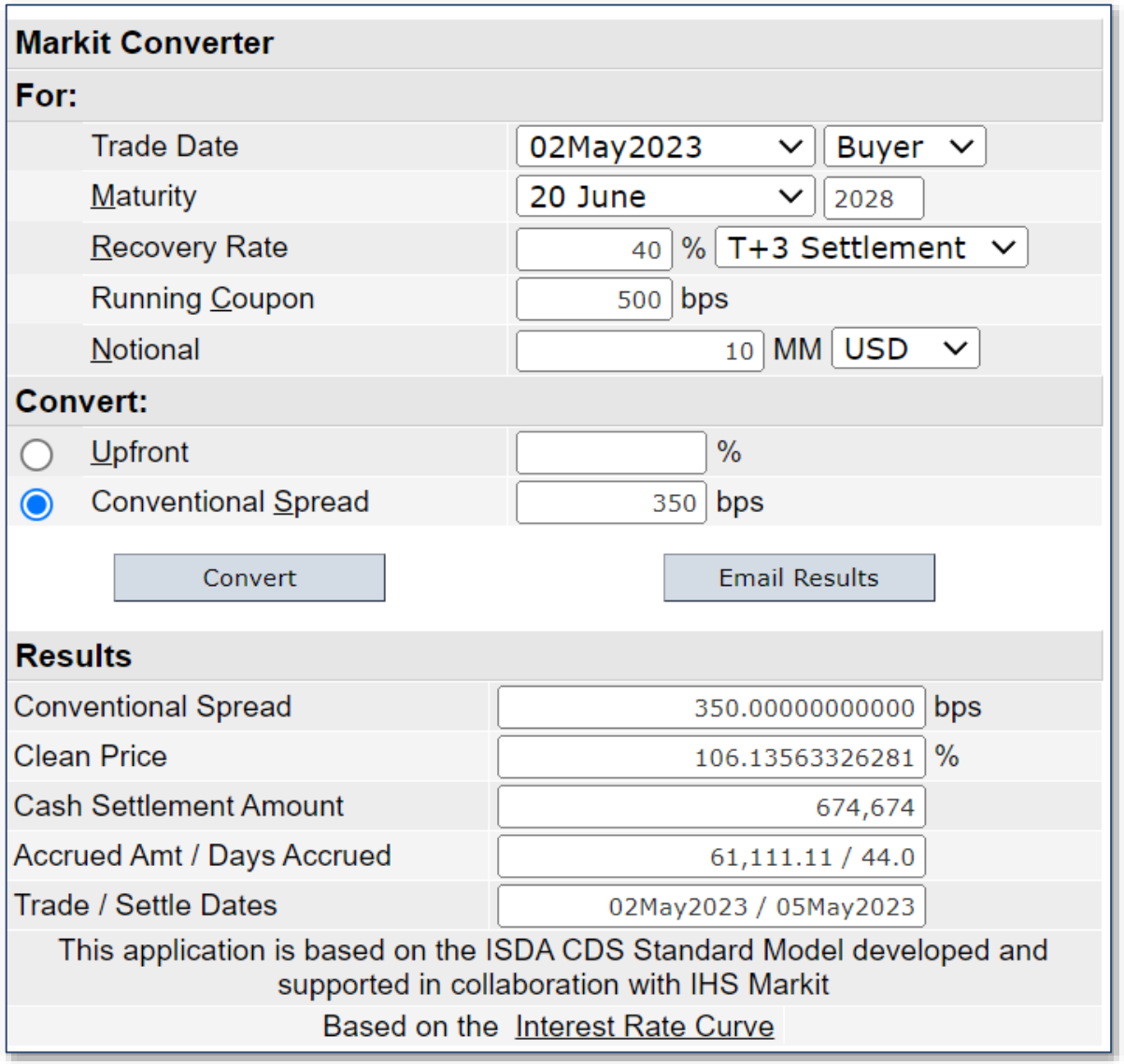
- Standard recovery rates also apply:

	Tier of Debt	
	Senior	Subordinated
Asian Corporate & Sovereign CDS	40%	20%
Australian Corporate & Sovereign CDS	40%	20%
Eastern & Central European Corporate & Sovereign CDS	25%	25%
European Corporate CDS	40%	20%
Japanese Corporate & Sovereign CDS	35%	15%
Latin American Corporate & Sovereign CDS	25%	25%
Middle Eastern Corporate & Sovereign CDS	25%	25%
New Zealand Corporate & Sovereign CDS	40%	20%
North American Corporate & Sovereign CDS	40%	20%
Western European Sovereign CDS	40%	20%

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ISDA CDS Standard Model

- The ISDA CDS Standard Model performs the up-front payment calculations and is available:
  - On Bloomberg (CDSW page)
  - As an Excel add-in
  - Online (e.g. Markit)



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Recovery Rates By Seniority

Average corporate debt recovery rates measured by trading prices

	Issuer-weighted recoveries			
Priority position	**2022 with Russian defaults**	**2022 without Russian defaults**	**2021**	**1983-2022**
1st Lien Bank Loan	69.8%	69.8%	66.4%	65.7%
2nd Lien Bank Loan	54.7%	54.7%	34.1%	33.3%
Sr. Unsecured Bank Loan				46.3%
1st Lien Bond	60.5%	60.5%	65.9%	55.0%
2nd Lien Bond	64.7%	64.7%	98.1%	44.2%
Sr. Unsecured Bond	31.1%	35.1%	48.7%	37.7%
Sr. Subordinated Bond			17.0%	31.0%
Subordinated Bond	18.2%	24.8%		32.0%
Jr. Subordinated Bond				22.3%

Source: Moody's – Annual Default Study, Apr 2023

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Example CDS Quotations

CDS Spread Increases		CDS Spreads		
Issuer	Senior Ratings	Apr. 26	Apr. 19	Spread Diff
Casino Guichard-Perrachon SA	Caa2	9,585	8,230	1,355
Vedanta Resources Limited	Caa2	2,759	2,519	240
Boparan Finance plc	Caa3	2,124	2,000	124
Novafives S.A.S.	Caa2	1,162	1,055	107
TK Elevator Holdco GmbH	Caa1	517	455	62
Renault S.A.	Ba2	312	256	55
CPI Property Group	Baa3	649	596	53
Carnival plc	B3	1,044	994	50
Picard Bondco S.A.	Caa1	723	673	50
Garfunkelux Holdco 3 S.A.	Caa2	1,389	1,341	48

CDS Spread Decreases		CDS Spreads		
Issuer	Senior Ratings	Apr. 26	Apr. 19	Spread Diff
United Group B.V.	Caa1	944	1,181	-237
Trinseo Materials Operating S.C.A.	B2	854	897	-43
Nidda Healthcare Holding GMBH	Caa3	576	600	-24
Stagecoach Group Limited	Baa3	220	239	-19
Ardagh Packaging Finance plc	Caa1	791	808	-17
Iceland Bondco plc	Caa2	1,051	1,068	-17
Hamburg Commercial Bank AG	A3	157	169	-12
Nationwide Building Society	A1	71	82	-11
ASML Holding N.V.	A2	40	51	-11
Sappi Papier Holding GmbH	Ba2	314	325	-11

Source: Moody's – Market Data Highlights, 27 Apr 2023

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Top Forty Reference Entities Traded

1	People's Republic of China	15	Credit Suisse Group AG	29	Barclays plc
2	Federative Republic of Brazil	16	Verizon Communications Inc.	30	Lloyds Banking Group plc
3	United Mexican States	17	Electricite De France	31	Jaguar Land Rover Automotive plc
4	Republic of South Africa	18	AT&T Inc.	32	Basf SE
5	Republic of Turkey	19	United States of America	33	The Boeing Company
6	Republic of Korea	20	Volkswagen AG	34	Vodafone Group plc
7	Republic of Indonesia	21	Airbus SE	35	Glencore International AG
8	Republic of Chile	22	Telefonica, S.A.	36	Ford Motor Company
9	Republic of Colombia	23	British Telecommunications plc	37	United Kingdom
10	Kingdom of Saudi Arabia	24	Enel spa	38	Bank of America Corporation
11	Republic of Italy	25	French Republic	39	JPMorgan Chase & Co.
12	Malaysia	26	Deutsche Bank AG	40	Mercedes-Benz Group AG
13	Republic of The Philippines	27	BP plc		
14	Republic of Peru	28	Telecom Italia spa		

Source: DTCC Index Roll Report, 27 Aug 2022 through 24 Feb 2023

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Sovereign Reference Entities

Sovereign CDS differ from corporate CDS in key aspects:

- Pre-default restructuring is very common in the context of sovereigns
  - 2003 ISDA Definitions had to close 1999 ISDA Definitions loophole concerning 'mandatory' exchanges
- Bankruptcy **not applicable** to sovereigns
  - A sovereign cannot technically speaking go bankrupt
    - Cannot be dissolved after defaulting or wind itself up
  - Sovereigns are in theory only subject to their own laws and treaty obligations
    - Not subject to insolvency proceedings in foreign bankruptcy courts
    - Impossible to appoint an administrator, conservator, receiver or bankruptcy trustee
- By contrast standard waivers of sovereign immunity in most bonds, as well as legal precedent, mean that sovereigns can be taken to court for failure to pay or restructuring
  - Remedies can be enforced against the sovereign's assets held internationally and can freeze payment systems
- Applicable credit events:
  - **Repudiation/Moratorium** event is defined as both:
    - A Repudiation or a Moratorium announcement by a government authority or authorised officer; and
    - A subsequent Failure to Pay or Restructuring in any amount of obligations
  - **Failure to Pay** event is the same as for corporate CDS
  - **Restructuring** usually unmodified R but Mod-R applies to Australian and New Zealand sovereign CDS as standard
  - **Obligation Acceleration** is only standard for Latin American sovereigns

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Points Up-Front

- Initially used for high-yield (risky) names
- If high-risk names were quoted using running premiums, protection sellers would face excessive risk, because:
  - No matter how high the running premium quoted…
  - Following an early default, very little premium would actually be accrued.
- Such names are therefore quoted **points up-front**
  - A non-refundable up-front payment is made – these are the points up-front
  - Then 500bp is paid as premium running
  - Example: 27 + 500 means:
    - 27% of notional is paid as points up-front
    - 500bp of notional is paid as a running premium
- Table compares premiums for a 5yr CDS:
  - Quoted as a CDS with all premium running
  - Quoted points-up front with 500 bp p.a. running
  - Assumptions:
    - Level yield curve at 5% p.a.
    - Level probability of default curve
    - Recovery rate of 40%
- Now Points Up-Front an essential feature of all CDS trading with fixed running premiums (e.g. under SNAC)

Probability of Default	Running Premium (bp)	Points Up-Front (%)
4%	500	0.0%
5%	632	4.4%
6%	766	8.5%
7%	903	12.3%
8%	1,043	15.7%
9%	1,187	18.9%
10%	1,333	21.8%
11%	1,483	24.5%
12%	1,636	27.0%
13%	1,793	29.2%
14%	1,953	31.3%
15%	2,118	33.2%
16%	2,286	35.0%
17%	2,458	36.6%
18%	2,634	38.1%

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Points Up-Front and Price

For high-yield names, the up-front payment becomes the dominant factor

- The higher the CDS premium…
- … the higher the up-front payment

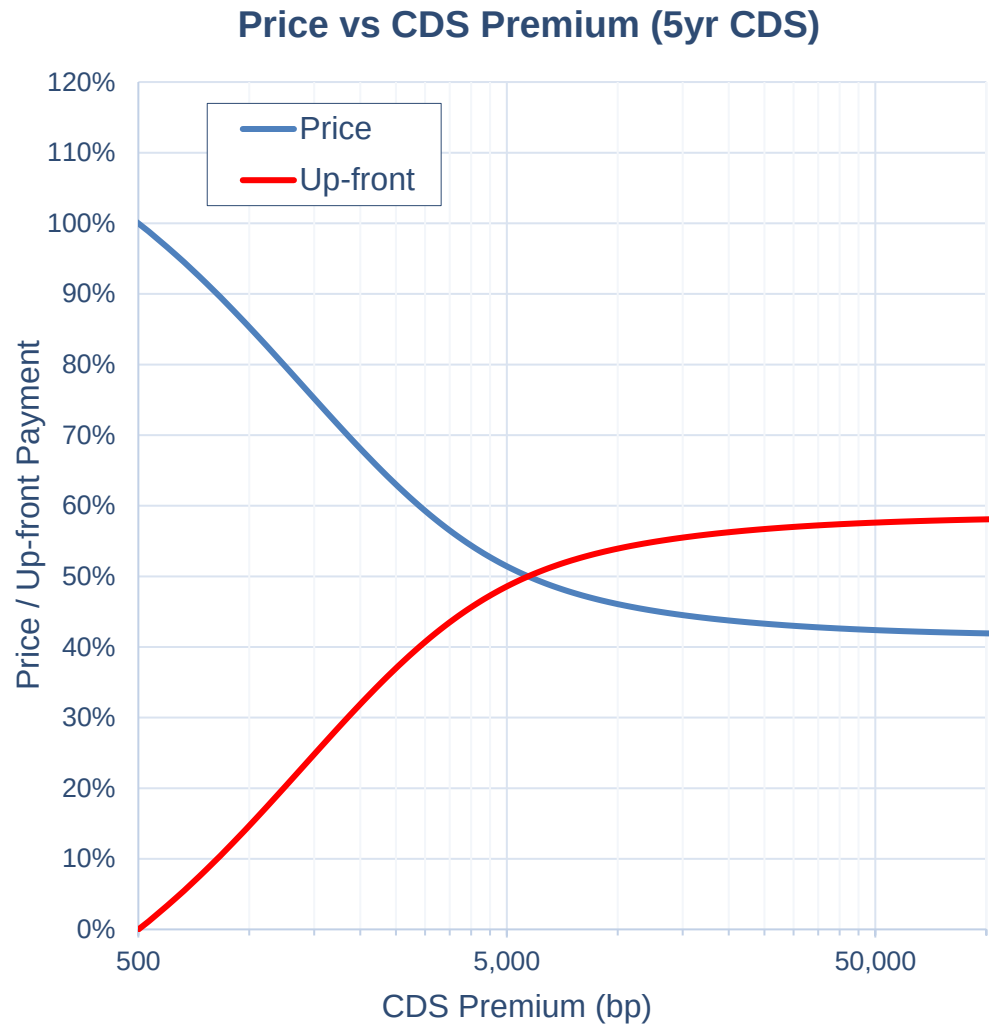
Markets often quote the up-front payment as a price

Price = 100% − Up-front payment

The chart opposite shows the relationship between

- CDS Premium
- Up-front payment / Price

… assuming a 40% recovery rate



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Recovery Products

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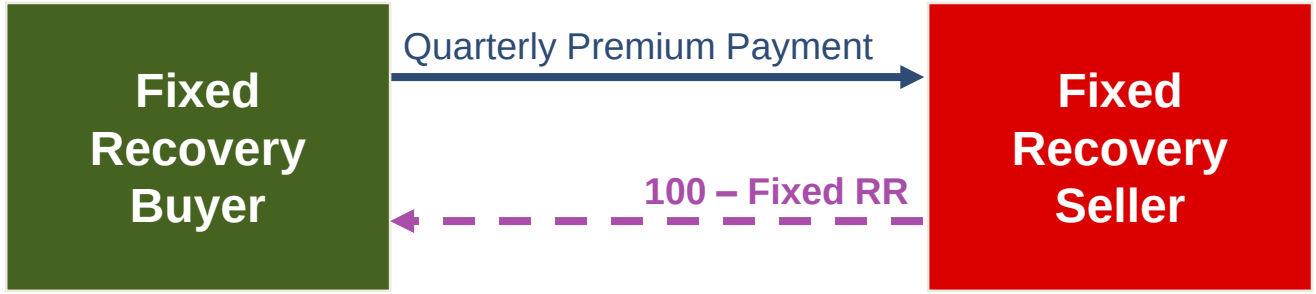
Recovery Products

- **Recovery Products** allow traders and investors to price and trade their views about the recovery rate separately
- The main recovery products are:
  - Recovery Locks
  - Recovery Swaps
- A complementary product is:
  - Fixed Recovery CDS
- … which allows traders and investors to price and trade their views about the probability of default separately

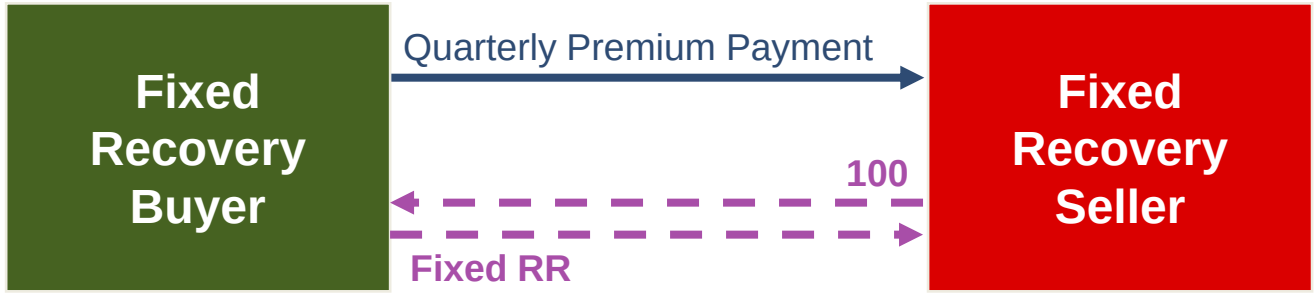
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Fixed Recovery CDS

- A **Fixed Recovery CDS** (aka Digital Default Swap) fixes the recovery rate at inception:



- Conceptually, on the occurrence of a credit event, the buyer of a fixed recovery CDS:
  - Receives 100
  - Pays the fixed recovery rate



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Recovery Locks

- A **Recovery Lock** is a forward transaction on the eventual recovery rate
  - No running premium
  - No up-front payment
  - If a credit event occurs, parties exchange the actual recovery rate for a fixed recovery rate



- As there is no premium, the fixed recovery rate is set to the market implied recovery rate
- Buyer of recovery lock is long recovery – makes money when RR rises
- Recovery locks are the market standard for trading recovery

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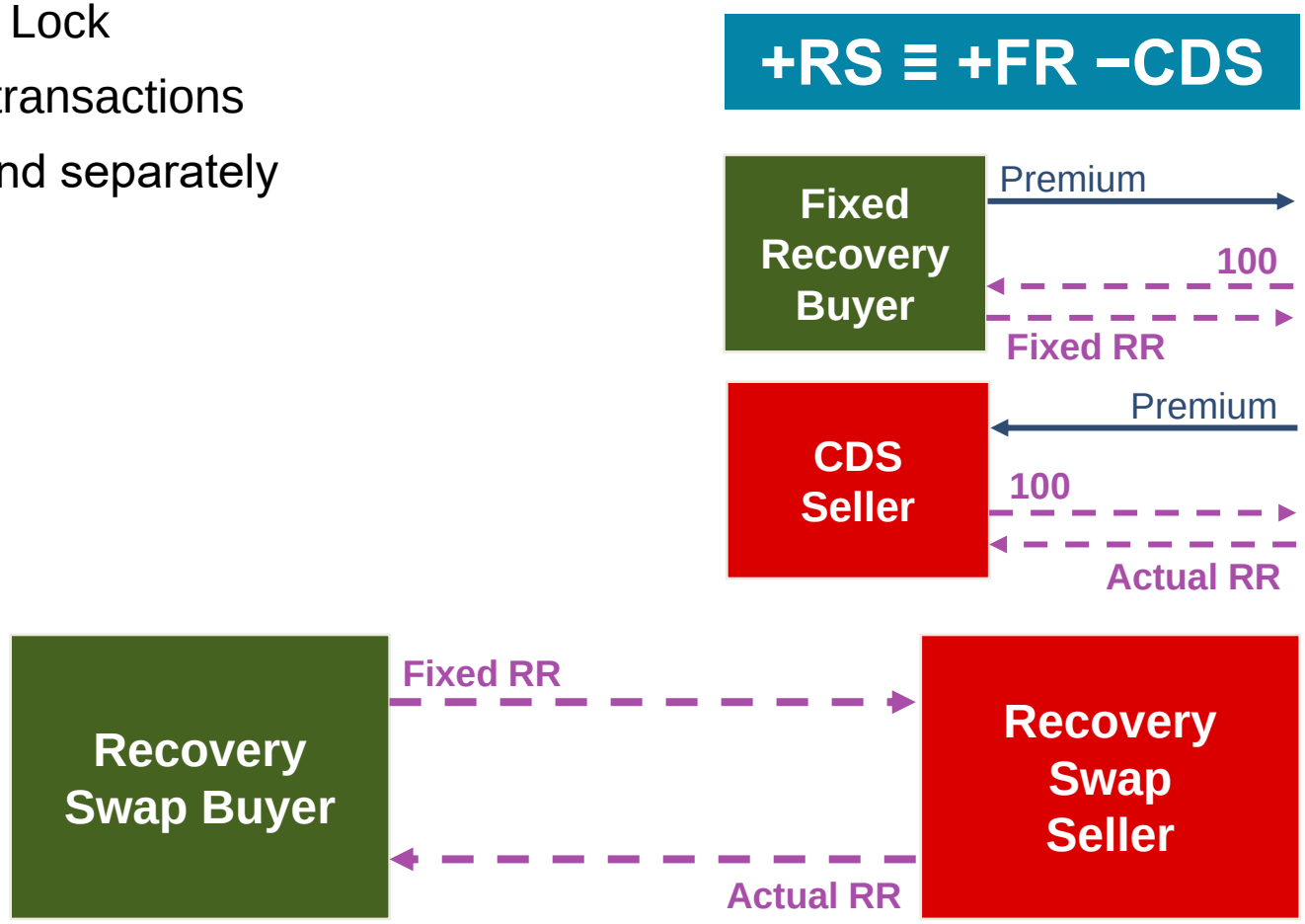
Recovery Swaps

A Recovery Swap is similar to a Recovery Lock

- …but is executed as two back-to-back transactions
- … and the two contracts can be unwound separately

The Recovery Swap buyer:

- Buys a Fixed Recovery CDS
  - Pays the premium
  - On a credit event:
    - Receives 100
    - Pays the fixed recovery rate
- Sells a standard CDS
  - Receives the premium
  - On a credit event:
    - Pays 100
    - Receives the actual recovery rate



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Long and Short Recovery

- A vanilla CDS is **short recovery** because:
  - If recovery rates go up, the amount received by the protection buyer goes down
  - The protection buyer therefore loses
- Recovery Locks and Recovery Swaps are **long recovery** because:
  - If recovery rates go up, the buyer receives a higher payment
  - The recovery lock (swap) buyer therefore gains

Long Recovery	Short Recovery
−CDS	+CDS
+RL	−RL
+RS	−RS

- In the ISDA standard documentation for recovery products:
  - Recovery Lock Buyer is called the protection seller, because he receives the actual recovery rate
  - Recovery Lock Seller is called the protection buyer, because he pays the actual recovery rate

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Recent Events

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Recent Events – ISDA 2014 Definitions

Key Areas of Changes to the Definitions:

- 'Government Intervention' (GI) – credit event triggered by governmental action in relation to the obligations of a financial entity, leading to:
  - Reduction in interest or principal payable
  - Postponement of payments
  - Subordination of obligations
  - Expropriation
  - Mandatory cancellation
- Asset Package Delivery (APD)
  - Following a Restructuring or a Governmental Intervention, provisions allowing the delivery of proceeds other than 'Deliverable Obligations' irrespective of their form…
  - … provided they are resulting from bailed-in obligations or restructured Reference Obligations that were deliverable immediately prior to the occurrence of the Credit Event
- Standard Reference Obligation (SRO)
  - For more frequently-traded Reference Entities
  - An obligation will be selected to be the SRO and published on the SRO List
  - This SRO will then be used as the reference obligation for all market standard CDS contracts on that reference entity and seniority level

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Recent Events – ISDA 2014 Definitions (cont'd)

- Qualifying Guarantees
  - Expansion of the scope of guarantees that pass the test of being a 'good' guarantee under the CDS Definitions and can therefore be hedged with a CDS contract
- Successors
  - Debt movement without corporate event is sufficient for the determination of Successor
  - Debt transfers can now be considered over a period of time (Step Plan)
  - Creation of the concept of Universal Successor that can be determined outside the 90-Day look back period
- Euro exit / currency redenomination
  - Provisions so that a debt change of currency following a Euro exit would not necessarily trigger a restructuring
- Restructuring Mechanics
  - Updates to Exercise Cut-off Date in case of No Auction Announcement and to Movement Option Cut-off Date
  - Removal of Enabling Obligations
  - Triggering without Credit Event Notice of transactions under 'Old' Restructuring

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Central CounterParty – Timeline

Establishment of a CCP (Central CounterParty) / Derivatives Clearing Service for CDS and index trades

- **Dec 2008:** NYSE.Liffe / LCH.Clearnet service launched for clearing iTraxx trades – but suspended service on 29th July 2009 as the service failed to gain traction
- **Mar 2009:** ICE Trust (Intercontinental Exchange, Atlanta) launched
- **Dec 2009:** CME launches Credit Derivative clearing operation
- **Mar 2010**: LCH.Clearnet clears first index contracts
- **Jul 2011:** ICE Trust transitions to ICE Clear Credit
  - An CFTC-regulated Derivatives Clearing Organization
  - A SEC-regulated Securities Clearing Agency
- **2 May 2023:** Cumulative gross notional volumes cleared since inception:

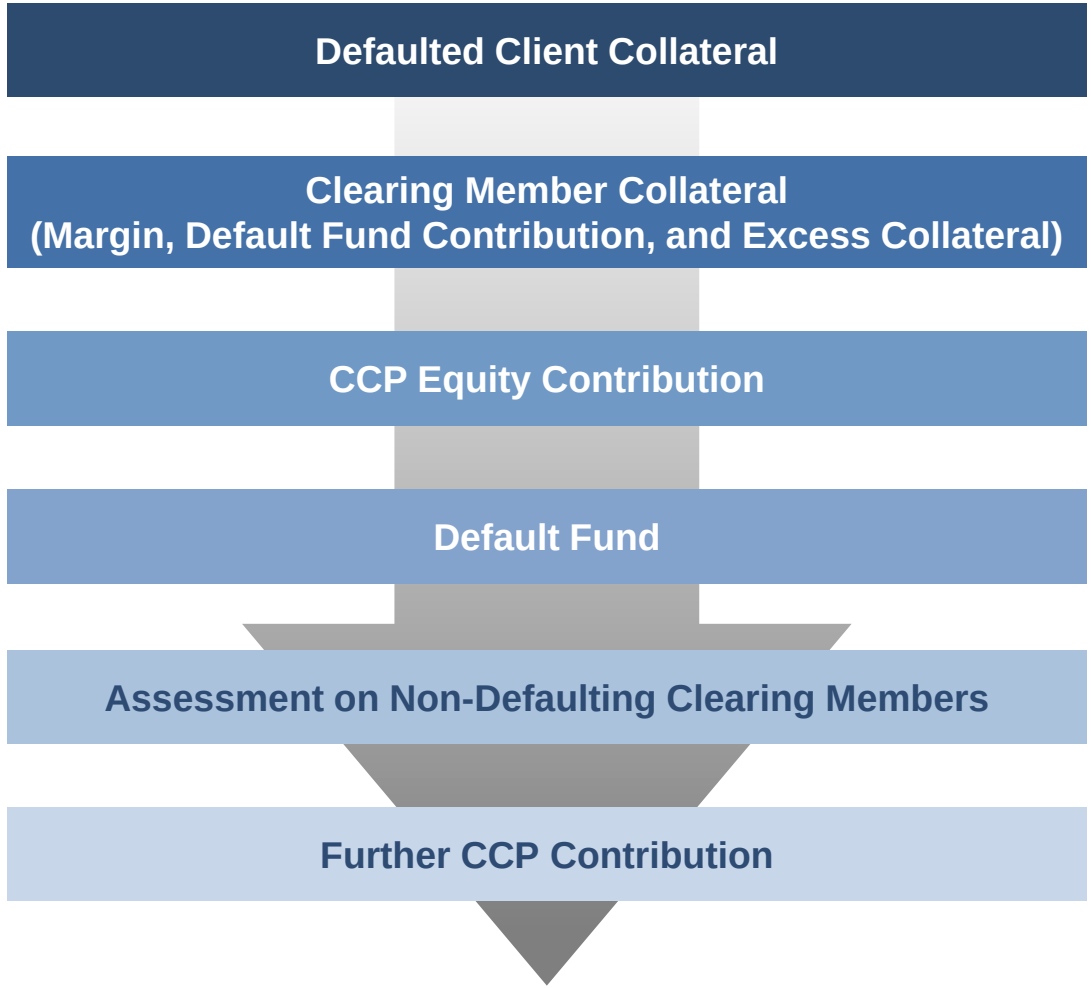
ICE Clear Credit	$138 trn	190 indices
	$8 trn	549 single-name CDS
	$5 trn	40 sovereigns
ICE Clear Europe	€30 trn	130 indices
	$4 trn	190 single-name CDS
	$0.7 trn	7 sovereigns

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Central CounterParty – Waterfall of Resources

Waterfall of CCP resources:

- Defaulting member's variation and initial margin
- Defaulting member's default fund contribution
- Juniorised default fund contributions
- CCP's own capital and reserves
- Surviving members' default fund contributions
- Pre-agreed top-up contributions to the CCP fund
- Margin accounts of every remaining member firm that is a net receiver of margin
  - … achieved by taking a haircut on the variation margin received, according to a set formula

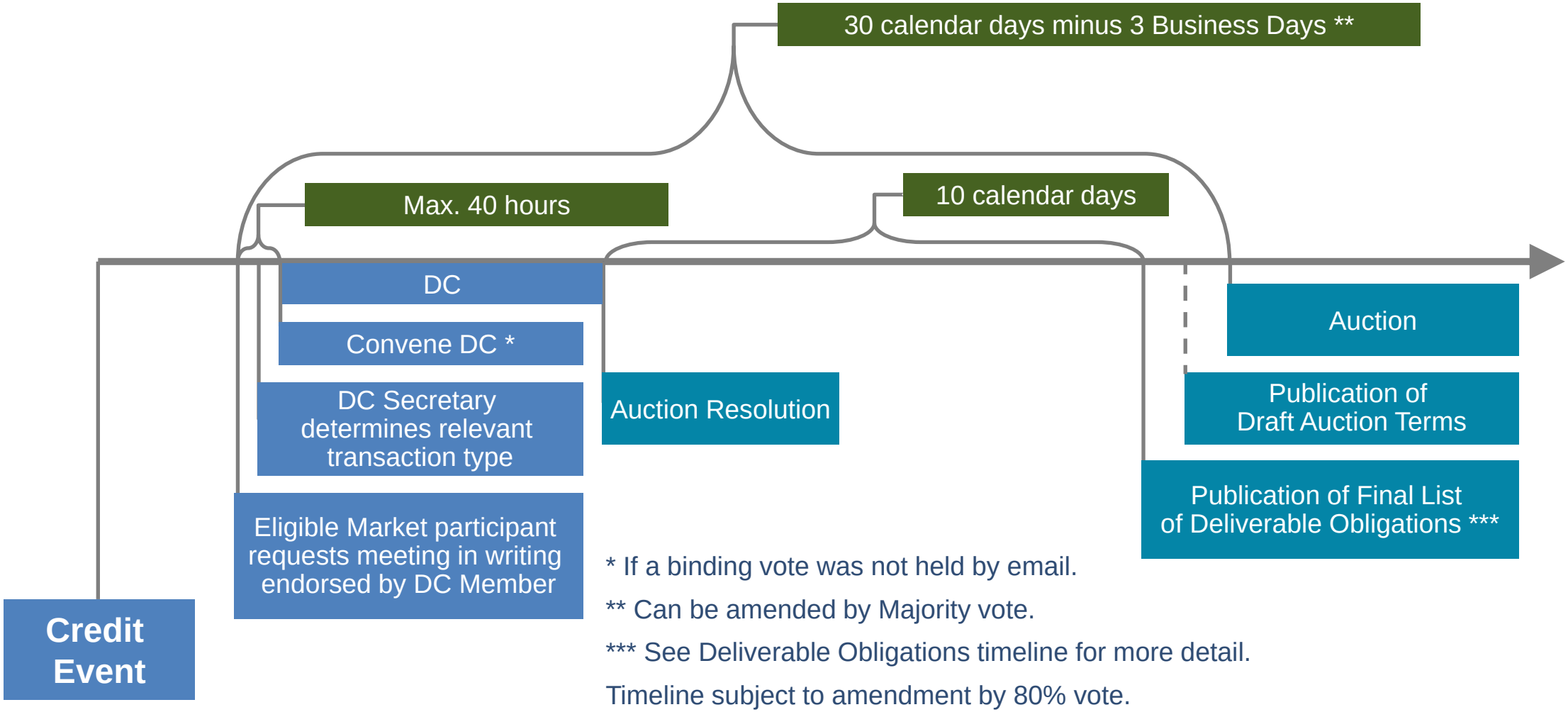


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How Auction Settlement Works

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Credit Event and Auction Timeline



* If a binding vote was not held by email.

** Can be amended by Majority vote.

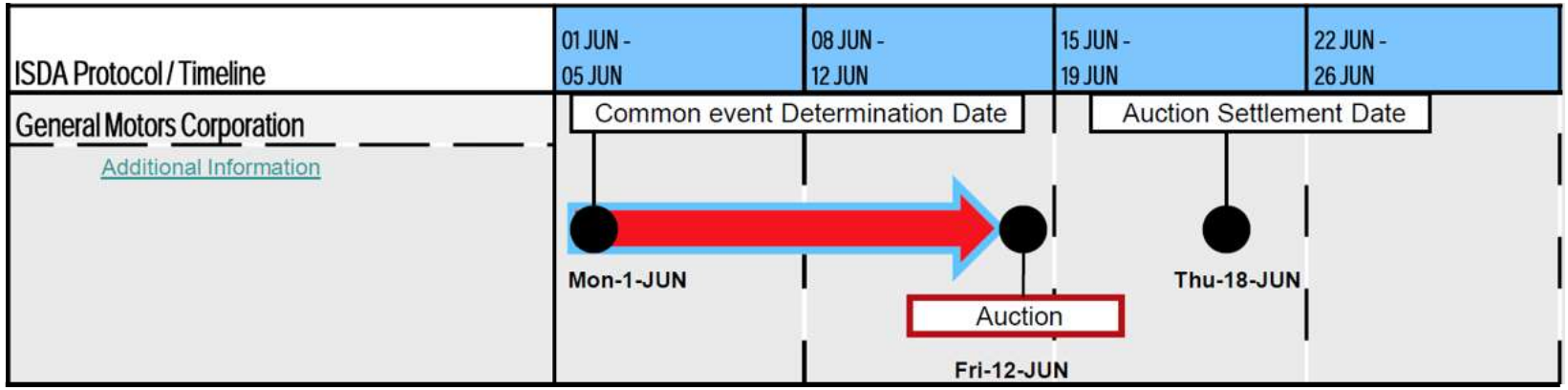
*** See Deliverable Obligations timeline for more detail.

Timeline subject to amendment by 80% vote.

Source: ISDA

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General Motors – Auction Timeline



- 1 June 2009 GM files for bankruptcy under Chapter 11
- 1 June 2009 DC votes 15:0 that:
  - A Bankruptcy Credit Event has occurred to General Motors Corporation
  - The Credit Event occurred on 1 June 2009
  - The DC Secretary received a request to convene the DC on 1st June
  - An auction should be held
- 1 June 2009 Initial List of Deliverable Obligations Published
- 4 June 2009 Supplemental List of Deliverable Obligations Published
- 8 June 2009 Final List of Deliverable Obligations Published
- 8 June 2009 Final Auction Terms Published
- 10 June 2009 List of Participating Bidders Published
- 10 June 2009 Updated Final List of Deliverable Obligations Published

Participating Bidders

- Bank of America / Merrill Lynch
- Barclays
- BNP Paribas
- Citibank
- Credit Suisse
- Deutsche Bank AG
- Goldman Sachs
- HSBC Bank USA, National Association
- JPMorgan Chase Bank, N.A.
- Morgan Stanley
- The Royal Bank of Scotland
- Société Générale
- UBS

Source: ISDA

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General Motors – Final List of Deliverable Obligations

CUSIP/ISIN	Issuer Name	Coupon	Maturity	Currency
370442691	General Motors	1.50%	June 1, 2009	USD
370442188	General Motors	7.200%	January 15, 2011	USD
37045EAS	General Motors	9.450%	November 1, 2011	USD
XS0171942757	General Motors	7.25%	July 3 , 2013	EUR
370442BS	General Motors	7.13%	July 15, 2013	USD
XS0171922643	General Motors, Nova Scotia Finance Company	8.375 %	December 7, 2015	GBP
370442AU	General Motors	7.700%	April 5, 2016	USD
370442AJ	General Motors	8.80%	March 1, 2021	USD
370442AIN	General Motors	9.400%	July 15, 2021	USD
37045EAG	General Motors	9.40%	July 15, 2021	USD
XS0I7190 8063	General Motors, Nova Scotia Finance Company	8.875 %	July 10, 2023	GBP
370442BW	General Motors	8.250%	July 15, 2023	USD
370442AV	General Motors	8.100%	June 15, 2024	USD
370442AR	General Motors	7.400%	September 1, 2025	USD
370442AZ	General Motors	6.75%	May 1, 2028	USD
370442741	General Motors	4.500%	March 6, 2032	USD
370442733	General Motors	5.250%	March 6, 2032	USD
XS0171943649	General Motors	8.375%	July 5, 2033	EUR
370442BT	General Motors	8.375%	July 15, 2033	USD
370442717	General Motors	6.25%	July 15, 2033	USD
370442AT* *	General Motors	0%/7.75%	March 15, 2036	USD

*** The accredited amount of this deliverable obligation is 56.521153% of the face amount. Parties should note this when submitting Limit Order Submissions. Source: ISDA*

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Auction Settlement

- Following the '2009 Supplement', **Auction Settlement** is now the market standard and is 'hard-wired' into CDS documentation
- The Auction is a two-step single-price auction…
  - Initial Auction Period in which Participating Bidders submit:
    - Bid/offer prices
    - Physical Settlement Requests
  - The Initial Auction determines:
    - The Inside Market Mid-point
    - Adjustment Amount penalties if Participating Bidders submit prices which are either too low or too high
    - The Open Interest for executing trades in Deliverable Obligations
  - The Final Auction determines:
    - The Final Auction Price
- The following pages describe the auction process in detail…

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Auction Settlement – Initial Auction Period

Each participating dealer is expected to:

- Contribute executable two-way prices, known as ‘inside market orders’
  - Such orders must be in the inside-market quotation size (e.g. $5 million of face value)…
  - ... in increments of 0.125%, and within the maximum inside-market quotation spread (e.g. 2%)
- Place a ‘dealer physical settlement request’
  - A proprietary market order to buy or sell
  - Must be in the same direction as, and not in excess of, the market position (i.e. exposure) faced by the dealer to deliveries following the credit event.
- Place any valid ‘customer physical settlement requests’ it has received from customers with whom it has a trading relationship

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Auction Settlement – Initial Auction Period (cont'd)

- Initial auction period lasts 15 minutes
- Following the conclusion of the initial auction period, the administrators will publish:
  - The **inside market midpoint**
    - A representative level of the executable two-way 'inside market orders' placed by all participating dealers
  - The size and direction of the **open interest**, the difference between:
    - (a) the sum of all physical settlement buy requests, and
    - (b) the sum of all physical settlement sell requests
  - The details of any 'adjustment amounts' due from individual dealers

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Auction Settlement – Initial Auction Example (cont'd)

Inside Market Orders

Dealer	Bid	Ask	Dealer
The Royal Bank of Scotland	12.000	10.500	Credit Suisse
BNP Paribas	11.000	10.750	JPMorgan Chase Bank, N.A.
HSBC Bank	11.000	11.000	Goldman Sachs
UBS	10.500	11.000	Morgan Stanley
Bank of America / Merrill Lynch	10.250	11.750	Citibank
Societe Generale	10.250	12.000	Barclays
Barclays	10.000	12.000	Deutsche Bank AG
Deutsche Bank AG	10.000	12.250	Bank of America / Merrill Lynch
Citibank	9.750	12.250	Societe Generale
Goldman Sachs	9.000	12.500	UBS
Morgan Stanley	9.000	13.000	BNP Paribas
JPMorgan Chase Bank, N.A.	8.750	13.000	HSBC Bank
Credit Suisse	8.500	14.000	The Royal Bank of Scotland

- Sort bids in descending order, offers in ascending order
- Identify crossing or touching bids

Inside Market Midpoint

Dealer	Bid	Ask	Dealer
Inside Market Midpoint = 11.000			
UBS	10.500	11.000	Morgan Stanley
Bank of America / Merrill Lynch	10.250	11.750	Citibank
Societe Generale	10.250	12.000	Barclays
Barclays	10.000	12.000	Deutsche Bank AG
Deutsche Bank AG	10.000	12.250	Bank of America / Merrill Lynch
Citibank	9.750	12.250	Societe Generale
Goldman Sachs	9.000	12.500	UBS
Morgan Stanley	9.000	13.000	BNP Paribas
JPMorgan Chase Bank, N.A.	8.750	13.000	HSBC Bank
Credit Suisse	8.500	14.000	The Royal Bank of Scotland

- Remove crossing or touching bids and identify best half (rounded up if an odd number)
- **Inside Market Midpoint** is simple average of best half, rounded to nearest 1/8th

General Motors Corporation
12th June 2009

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Auction Settlement – Initial Auction Example (cont'd)

Physical Settlement Requests

Dealer	Bid / Offer	Size $/m
Credit Suisse	Offer	407.000
HSBC Bank	Offer	0.000
Morgan Stanley	Offer	18.000
UBS	Offer	40.000
BNP Paribas	Bid	9.900
Bank of America / Merrill Lynch	Bid	21.146
Barclays	Bid	86.779
Citibank	Bid	126.000
Deutsche Bank AG	Bid	108.635
Goldman Sachs	Bid	144.200
JPMorgan Chase Bank, N.A.	Bid	237.438
The Royal Bank of Scotland	Bid	240.000
Societe Generale	Bid	20.000

- The **open interest** of $529.098 million to buy is the difference between:
  - The total requests to buy of $994.098 million, and …
  - … the total requests to sell of $465 million

General Motors Corporation
12th June 2009

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Auction Settlement – Initial Auction Example (cont'd)

Crossing Quotes

Dealer	Bid	Ask	Dealer
Bank of America / Merrill Lynch	10.250	12.250	Bank of America / Merrill Lynch
Barclays	10.000	12.000	Barclays
BNP Paribas	11.000	13.000	BNP Paribas
Citibank	9.750	11.750	Citibank
Credit Suisse	8.500	10.500	Credit Suisse
Deutsche Bank AG	10.000	12.000	Deutsche Bank AG
Goldman Sachs	9.000	11.000	Goldman Sachs
HSBC Bank	11.000	13.000	HSBC Bank
JPMorgan Chase Bank, N.A.	8.750	10.750	JPMorgan Chase Bank, N.A.
Morgan Stanley	9.000	11.000	Morgan Stanley
The Royal Bank of Scotland	12.000	14.000	The Royal Bank of Scotland
Societe Generale	10.250	12.250	Societe Generale
UBS	10.500	12.500	UBS

Adjustment Amounts

Dealer	Bid	Ask	Dealer
The Royal Bank of Scotland	12.000	10.500	Credit Suisse
BNP Paribas	11.000	10.750	JPMorgan Chase Bank, N.A.
HSBC Bank	11.000	11.000	Goldman Sachs

- If open interest is to buy, so that market prices should rise, low offers (relative to bids **and** IMM) pay adjustment amount

In this example:

- JPMorgan would pay ISDA: 0.25% × $5m = $12,500 (low offer of 10.750)
- Credit Suisse would pay ISDA: 0.50% × $5m = $25,000 (low offer of 10.500)

General Motors Corporation
12th June 2009

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Auction Settlement – Final Auction Period

- The final auction takes place 90 minutes after the end of the initial auction
- Final Auction typically lasts a further 15 minutes
- In the final auction, each participating dealer may submit limit orders:
  - Only in the opposite direction to the 'open interest'
  - Several limit bids or limit offers may be submitted for the dealer's own trading account…
  - … or on behalf of customers who wish to place limit orders
  - The orders can be as defensive as desired but must not improve on the 'inside market midpoint' by more than 1%
  - Limit order volumes must be expressed in units of the standard order size (e.g. $1m)
- 'Inside market orders' already submitted in the initial auction
  - Remain executable, if in the opposite direction to the 'open interest'
  - Are discarded, if in the same direction
- If limit price of an executable inside market order is deemed to be away from fair value as indicated by the 'inside market midpoint':
  - Limit price is adjusted to equal the 'inside market midpoint'
  - Executable part of inside market is combined with submitted limit orders to form limit order book

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Auction Settlement – Final Auction Example

- The **final auction price** is determined thus:
  - The $529.098 million open interest to buy were matched with…
  - … $ 529.098 million of limit offers down to 12.500%
  - In the table opposite:
    - The PSR trades and the next 26 offers resulted in trades.
    - All the offers above 12.500 were unsuccessful.
- All **trades** are then settled at the final auction price of **12.500%**
  - The $994.098 million requests to buy
  - The $465.000 million requests to sell
  - The $529.098 million of filled limit offers
- UBS, Citi, and Credit Suisse each receive a *pro rata* allocation on their 12.500% marginal bids
  - UBS is allocated 95.15% of the $5m offered = $4.757m
  - Citi is allocated 95.15% of the $50m offered = $47.575m
  - Credit Suisse is allocated 95.15% of the $4m offered = $3.806m
- This matches all Physical Settlement Requests
- All **CDS** are **cash settled** using this final auction price

Dealer	Order type	Size $/mio	Limit
BNP Paribas	PSR Buy	9.900	
Bank of America / Merrill Lynch	PSR Buy	21.146	
Barclays	PSR Buy	86.779	
Citibank	PSR Buy	126.000	
Deutsche Bank AG	PSR Buy	108.635	
Goldman Sachs	PSR Buy	144.200	
JPMorgan Chase Bank, N.A.	PSR Buy	237.438	
The Royal Bank of Scotland	PSR Buy	240.000	
Societe Generale	PSR Buy	20.000	
Credit Suisse	PSR Sell	407.000	
HSBC Bank	PSR Sell	0.000	
Morgan Stanley	PSR Sell	18.000	
UBS	PSR Sell	40.000	
Goldman Sachs	Limit Offer	1.000	10.000
Credit Suisse	Adjusted Inside Offer	5.000	11.000
JPMorgan Chase Bank, N.A.	Adjusted Inside Offer	5.000	11.000
Goldman Sachs	Inside Market Offer	5.000	11.000
Morgan Stanley	Inside Market Offer	5.000	11.000
BNP Paribas	Limit Offer	11.960	11.000
Deutsche Bank AG	Limit Offer	5.000	11.000
Citibank	Inside Market Offer	5.000	11.750
Barclays	Limit Offer	25.000	11.875
Deutsche Bank AG	Limit Offer	25.000	11.875
Barclays	Inside Market Offer	5.000	12.000
Deutsche Bank AG	Inside Market Offer	5.000	12.000
Citibank	Limit Offer	60.000	12.000
Barclays	Limit Offer	25.000	12.000
Citibank	Limit Offer	50.000	12.000
Deutsche Bank AG	Limit Offer	25.000	12.000
Citibank	Limit Offer	50.000	12.125
Bank of America / Merrill Lynch	Inside Market Offer	5.000	12.250
Societe Generale	Inside Market Offer	5.000	12.250
Citibank	Limit Offer	50.000	12.250
Goldman Sachs	Limit Offer	25.000	12.250
Citibank	Limit Offer	50.000	12.375
Goldman Sachs	Limit Offer	25.000	12.375
UBS	Inside Market Offer	5.000	12.500
Citibank	Limit Offer	50.000	12.500
Credit Suisse	Limit Offer	4.000	12.500
Citibank	Limit Offer	75.000	12.625
Citibank	Limit Offer	130.000	12.750
Citibank	Limit Offer	75.000	12.875
Deutsche Bank AG	Limit Offer	25.000	12.875
BNP Paribas	Inside Market Offer	5.000	13.000
HSBC Bank	Inside Market Offer	5.000	13.000
Citibank	Limit Offer	50.000	13.000
Barclays	Limit Offer	50.000	13.000
Barclays	Limit Offer	25.000	13.000
Credit Suisse	Limit Offer	30.000	13.000

66 further offers…

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Introduction to CDS Pricing

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Principles of CDS Pricing – Understanding Asset Swaps

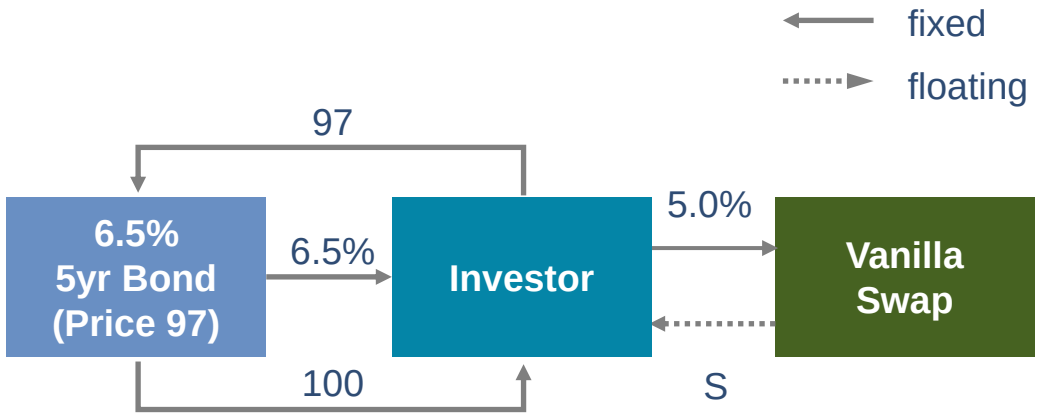
- An asset swap is a vanilla IRS which has been customised to match a specific fixed-interest bond
- The aims are:
  - To swap the bond cash flows from fixed to floating
  - For a *par asset* swap, to change the up-front investment to par
- The asset swap is a customised vanilla swap in which:
  - The fixed rate of the swap matches the fixed coupon of the bond
  - For a *par asset* swap, there is an up-front payment corresponding to the difference between the bond's purchase price and par
- An asset swap is therefore an off-market margin swap

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Asset Swaps – Example

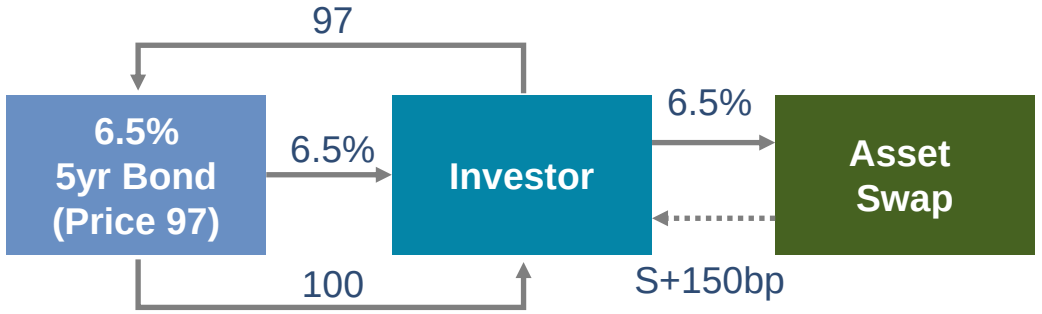
Step 1

- Investor buys bond
- Investor enters into vanilla swap
- Problem: fixed payments don't match



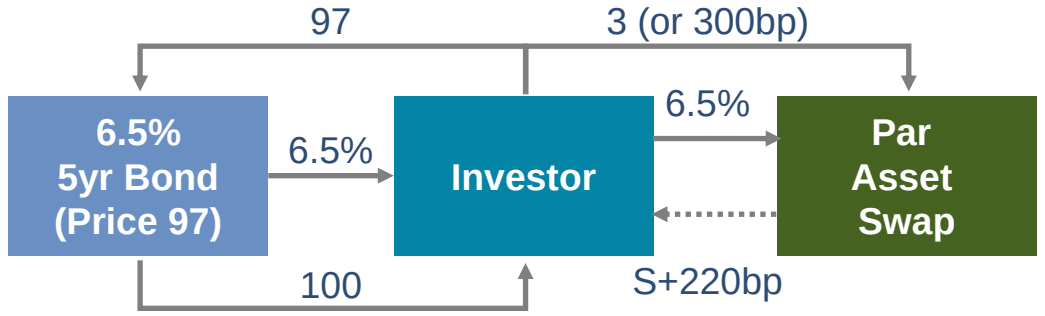
Step 2

- Swap rate adjusted to match bond coupon
- Floating rate adjusted accordingly



Step 3

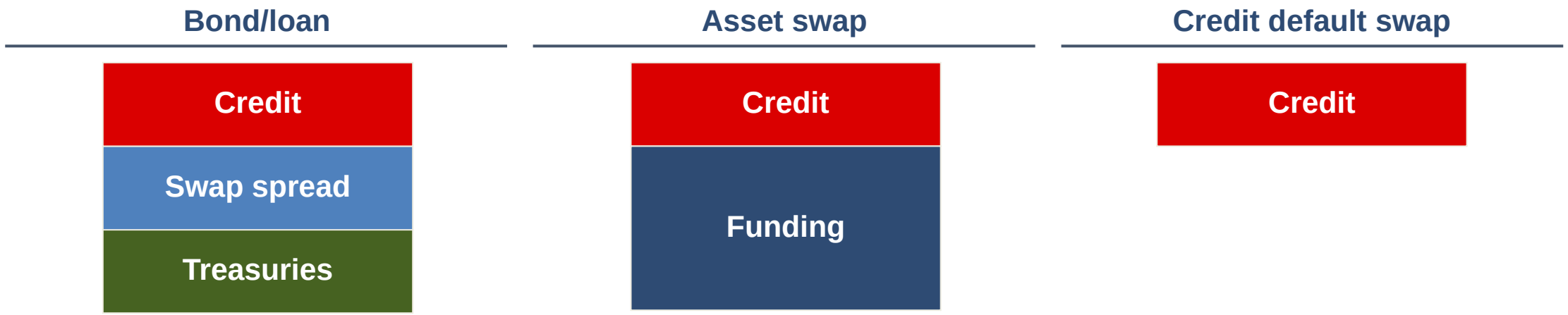
- Investor makes up-front payment to swap c'party
- Floating rate adjusted accordingly



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Principles of CDS Pricing – Asset Swap Approach

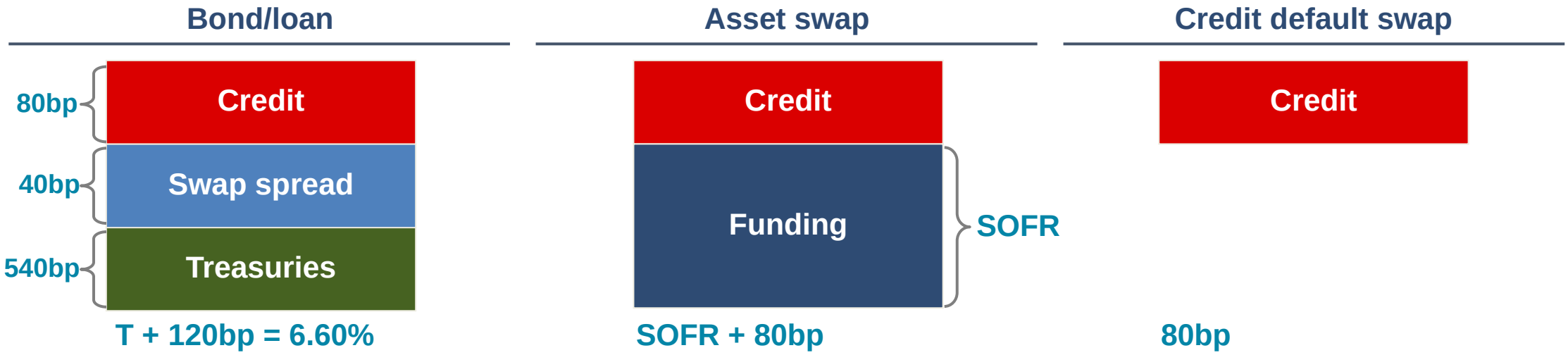
- A bond or loan contains interest rate risk, funding risk, and credit risk
- Credit default swaps isolate and transfer credit risk
- To determine the theoretical price of credit risk
  - Remove the interest rate risk by swapping the bond into a floating rate investment (asset swap)
  - Remove the funding cost (assume a funding of SOFR)
  - Credit risk remains, which gives you the theoretical price for credit protection



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CDS Pricing Example – Asset Swap Approach

- Bonds
  - 5yr Treasuries trade at 5.40%
  - XYZ 5yr 6.6% Bond trades at a yield of 6.60% (T+120bp)
- Asset Swap
  - 5yr Vanilla IRS trades at 5.80% (i.e. swap spread is 40bp)
  - 5yr Asset Swap package on XYZ 5yr 6.6% bond is quoted at S+80bp



In practice, more sophisticated pricing techniques are used. (See module on CDS pricing).

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